What changed since 7 November

From `63ab376` (Léonard, 7 November 20:58) to `b37cfa7` (8 November 17:12)

Eight commits on `feat/stage-a-foundations`. 37 files, about 3,100 lines added. Written for Léonard, so it says what moved, what broke, which numbers changed and what is still open.

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1. The short version

Stage A ran end to end on Engaging for the first time, in 8 minutes on a CPU node. Everything in `docs/corpora.md` now has a regenerated counterpart under `runs/analysis/stage_a_corpora/`, plus seven new tables that did not exist before. The results were then put through a verification pass against the code, the raw samples, the original papers and the lecture slides. That pass confirmed every arithmetic value and corrected six framing or definitional errors, three of which were in the first version of the check-in report and one of which is in the pipeline itself.

Three documents are new: the check-in 1 report, a literature review of the twelve papers on the reading list, and a Stage C task plan awaiting your validation. All three exist as Markdown in `docs/` and as PDF and Word in `docs/out/`.

Nothing in the experiment design changed. The stages, the corpora, the fixed recipe and the evaluation protocol are as you left them.

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2. Bugs found and fixed

Data paths split in two. The downloaders wrote under `NC_DATA_DIR` while every reader resolved `source:` against the repo root, so pointing the data at scratch made `build_cache.py` report a missing corpus for a file that had downloaded correctly. All five call sites now go through one resolver, `natural_compounds.paths.resolve_source`, with tests. This is why data can now live on scratch, which the cluster guide asks for.

Splits included unparseable rows. `scaffold_split` grouped the 11 BBBP SMILES RDKit cannot read under a pseudo-scaffold and placed them like any other group, so the frozen splits held rows no model can train on. They now belong to no subset, and `make_splits.py` says how many it dropped. **The BBBP split sizes changed: 1,631 / 203 / 205 instead of 1,640 / 205 / 205.** ESOL is unchanged at 902 / 112 / 114.

Downloads were serial and could not resume. Zenodo and the COCONUT bucket serve a single connection at about 100 KB/s from the cluster, while the pipe itself carries 4.7 MB/s; COCONUT took over an hour and the host dropped the connection twice. Each file now comes down as eight byte ranges, each range resumes where it stopped, and a file left short by an interrupted run is fetched again instead of being reported as a digest mismatch forever. The full download is now about four minutes.

The Slurm log directory did not exist, so the first job's output went nowhere. Created before submission and in `session.sh`.

`session.sh` uninstalled the training stack. It ran `uv sync --extra descriptors`, which is exact and removes anything else. It now syncs once with what Stage A needs.

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3. Numbers of yours that changed

Everything in `docs/corpora.md` reproduces exactly: cache counts, natural-product-likeness means, benchmark exposure. Two things differ, both because the statistic was redefined rather than recomputed.

Scaffold concentration. `scripts/corpus_report.py` counted the empty Murcko scaffold of acyclic

molecules as one scaffold category. For COCONUT that bucket holds 12.3 percent of molecules and sat at rank one, which made COCONUT look as scaffold-concentrated as PubChem. Concentration statistics are now computed over ring-containing molecules only. **PubChem, not COCONUT, is the most scaffold-concentrated corpus.**

A consequence worth knowing: `analyze_corpora.py` and `corpus_report.py` now use different denominators for `scaffolds_per_molecule`. Yours divides the scaffold count by all 20,000 molecules (COCONUT 0.528), the new one divides by the 17,535 that have a ring (COCONUT 0.602). Same numerator. Worth settling on one before the write-up; I did not change your script because its output is what `docs/corpora.md` quotes.

Benchmark exposure denominators. Exact matches are counted over unique canonical structures but were divided by raw row count. BBBP has 2,050 rows, 11 unparseable and 64 duplicated, so 1,975 unique structures. The shares moved by a fraction of a percent; the full-corpus headline figures (73.1 and 93.2 percent for PubChem, 36.1 and 55.6 for COCONUT) are unchanged and are now produced by a committed script, `full_corpus_exposure.csv`, rather than a one-off count.

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4. What is new in the pipeline

`scripts/corpus_report.py`, about 500 lines. Seven tables and two figures that `analyze_corpora.py` does not produce:

table	what it answers
properties.csv	16 descriptors per corpus, including Ertl's seven and the nitrogen/oxygen split
property_effects.csv	Cohen's d, Cliff's delta, probability of superiority and histogram overlap for all 78 corpus-pair comparisons
scaffolds.csv	counts, Pielou evenness, top-100 share, singleton shares, acyclic share
diversity.csv	pairwise similarity and sphere-exclusion coverage at two radii and two nested sample sizes
nearest_neighbour.csv	directional corpus-to-corpus similarity
task_exposure.csv	exact and near-duplicate benchmark matches, with every denominator stated
task_proximity.csv	per-benchmark distance to each corpus, the Stage B x-axis
full_corpus_exposure.csv	exact exposure against the whole caches

`slurm/stage-a.sbatch` runs all of Stage A on a CPU node. Every step is independent, so one failure cannot cost the rest, and every step skips work already on disk, so the job is safe to rerun and safe to preempt. It installs only what Stage A needs: nothing in it imports torch or chemprop, and syncing the training extra on a compute node pulled several GB of CUDA wheels at 0.2 MB/s, which cost one job two hours before I caught it.

`scripts/md2doc.py` builds the PDF and Word copies of any document in `docs/`, figures included. It needs a small separate environment, described in its docstring, because python-docx and fpdf2 have no business in the project dependencies.

Three new test files cover the path resolver, the byte-range splitter, the sphere-exclusion count, the scaffold handling and the pooled-SD effect size; the cache and split tests were extended. The suite is at 55 tests and runs in four seconds.

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5. What the verification pass changed

I ran the results through five independent checks (code review, independent recomputation from the raw

samples, comparison against the original papers, slide-by-slide checking of the lecture figures, and a line-by-line audit of every number in the report), then two adversarial reviews that tried to refute the two load-bearing claims.

Every arithmetic value reproduced, to six significant figures, from the raw 20,000-molecule samples using independent code. Both adversarial reviews failed to refute their target. What did not survive was a set of definitional and framing errors:

- The sphere-exclusion threshold was applied as a similarity cutoff where

ChemFM, the cited source, uses a distance threshold. At our radius three of four corpora saturate, so the "COCONUT decays 70 times faster" claim was a ceiling artefact and is withdrawn. **The finding survives on absolute numbers**: one COCONUT molecule in five in a 10,000-molecule sample has a near twin, against one in twenty for PubChem and under one in two hundred for either ZINC arm. Coverage is now reported at both radii, and the coarse one is directly comparable to ChemFM's published value.

- Proximity and near-duplicate counts used a 4,000-molecule reference while

exact matches used the whole 20,000, which made near-duplicate counts come out *below* exact-match counts. Both now use the whole condition. "More than 70 percent of the benchmarks are remote from ZINC" became about half.

- Four of the lecture figures in the report were overstated. The BACE

foundation-model advantage halves by 847 labels rather than vanishing; three rather than two pretrained models are worse than from-scratch on 45-molecule ESOL; the 0.008 pharma spread was the tightest of four assays, not a bound; and "more data, worse rank" is not monotone and reverses on MoleculeNet. The critical differences of 1.61 and 2.38 were confirmed exactly, including against the Nemenyi formula.

- Ertl's paper names no +1.0 cutoff, and his property model is a classifier

compared on enrichment, not a reproduction of the score. Both corrected here, in `docs/corpora.md` and in the script docstring.

Section 10 of the check-in report lists all of it.

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6. One new result worth your attention

CheMeleon's paper says its corpus was built by "randomly selecting 1 million molecules from PubChem". It was not. PubChem holds about 119 million compounds, so a uniform draw would contain under one percent of any benchmark; the file contains 73 percent of BBBP and 93 percent of ESOL. Ordering the corpus by row number, the first decile holds 1,260 of the 1,443 BBBP hits and 1,002 of the 1,041 ESOL hits, while three deciles hold no ESOL hit at all. Of twenty textbook molecules, nineteen are present where a uniform draw would contain 0.2 of them.

The released corpus is the identifier-ordered head of PubChem, which means the well-studied core. That is a property of the released checkpoint too: it has already seen 73 percent of the BBBP test split and 91 percent of ESOL's. It does not invalidate anything, but Stage B measures transfer to molecules the encoder mostly knows, and the write-up has to say so.

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7. Still open

- **The fine-tuning loop is still a stub.** It is the critical path: no

endpoint can be scored until it exists, and nothing in Stages B to E runs without it.

- **The implementation choice is still open.** `scripts/pretrain.py` drives

the original CheMeleon code, which is the only choice that keeps the released checkpoint comparable. Worth confirming rather than reopening.

- **The descriptor column set must be frozen** before the first pretraining

run, against the largest corpus on the curve. At n=500 there are 70 all-NaN columns; at n=4,000 there are 44. Otherwise the objective grows with the corpus and the size curve measures two things at once.

- **The two scaffold denominators** described in section 3.
- **Decisions in the check-in report, section 7**: whether `zinc_nplike`

stays and under what name, whether to add a property-matched PubChem arm, and how to handle benchmark overlap.

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8. Where things are

Results	runs/analysis/stage_a_corpora/ and its report/ subfolder
Data	~/orcd/scratch/natural-compounds-data/ (caches, conditions, splits, descriptors)
Check-in report	docs/check-in-1-report.md, PDF and Word in docs/out/
Literature review	docs/literature-review.md, same
Stage C plan	docs/stage-c-plan.md, same, awaiting validation
Paper sources	papers/README.md (PDFs are gitignored)

To reproduce everything from a clean checkout:

```
source slurm/session.sh
sbatch slurm/stage-a.sbatch
```